

Covariance-aware variant calling in single cell and spatial transcriptomics

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July 31, 2025

1 Introduction

In the following section we provide the derivation of a latent Gaussian process with a Binomial likelihood, to enhance the mutation calling in data with known covariance structure retrieved from single cell or spatial transcriptomics assay.

2 Latent Gaussian process with Binomial likelihood

2.1 Model definition

Consider a dataset collected from single cell RNA-seq, with N cells belonging to K known clones, V putative mutations. The observed variables are

$$\begin{array}{lcl} \mathbf{Y} = (\mathbf{y}_1, \dots, \mathbf{y}_V), \text{ with } \mathbf{y}_v = (y_{v1}, \dots, y_{vN}) & \left| \begin{array}{l} V \times N \\ V \times N \\ K \times 2 \\ N \\ N \times N \end{array} \right. & \begin{array}{l} \text{NV counts} \\ \text{DP counts} \\ \text{Edges (ancestor, child)} \\ \text{Cells assignments} \\ \text{Cells covariance} \end{array} \\ \mathbf{D} = (\mathbf{d}_1, \dots, \mathbf{d}_V), \text{ with } \mathbf{d}_v = (d_{v1}, \dots, d_{vN}) & & \\ \mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_K), \text{ with } \mathbf{x}_k = (x_{k1}, x_{k2}) & & \\ \mathbf{Z} = \{1, \dots, K\}^N & & \\ \Sigma & & \end{array}$$

We model the variant reads as a Binomial distribution

$$y_{vi} \sim \text{Binomial}(d_{vi}, \theta_{vi}) \quad (1)$$

The variable θ_{vi} corresponds to mutation v probability of success in cell i , belonging to clone k . To leverage the known covariance structure in the definition, we model the logit-transformed latent variable θ_v as a multivariate Normal distribution

$$\theta_{vi} = f(\gamma_{vi}) = \frac{1}{1 + e^{-\gamma_{vi}}} \quad (2)$$

$$\boldsymbol{\gamma}_v \sim \text{Normal}(\boldsymbol{\mu}_v, \boldsymbol{\Sigma}) \quad (3)$$

with $\boldsymbol{\Sigma}$ being the known covariance among cells. When this covariance encodes the phylogenetic distance among cells, we use the Ornstein-Uhlenbeck kernel, defined as follows for cells a and b

$$\Sigma_{ab} = \sigma^2 e^{-\lambda d(a,b)} \quad (4)$$

where $d(a, b) = (l_a + l_b - 2l_{\text{MRCA}_{ab}})$, with l_i being the branch length from root to node i , represents the phylogenetic distance among cells a and b [Revell, 2012].

Therefore, the latent variables of the model are

$$\begin{array}{lcl} \boldsymbol{\gamma} = (\boldsymbol{\gamma}_1, \dots, \boldsymbol{\gamma}_V), \text{ with } \boldsymbol{\gamma}_v = (\gamma_{v1}, \dots, \gamma_{vN}) & \left| \begin{array}{l} V \times N \\ V \times N \end{array} \right. & \begin{array}{l} \text{Logit-transformed prob. of success} \\ \text{Multivariate Normal mean vector} \end{array} \\ \boldsymbol{\mu} = (\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_V), \text{ with } \boldsymbol{\mu}_v = (\mu_{v1}, \dots, \mu_{vN}) & & \end{array}$$

2.2 Probability functions

We can write the **joint probability** of the model as

$$p(\mathbf{Y}, \boldsymbol{\gamma} | \mathbf{D}, \boldsymbol{\mu}) = \prod_{v=1}^V \left[\text{Normal}(\boldsymbol{\gamma}_v | \boldsymbol{\mu}_v, \boldsymbol{\Sigma}) \prod_{i=1}^N \text{Binomial}(y_{vi} | d_{vi}, f(\gamma_{vi})) \right] \quad (5)$$

and the **log-joint probability** as

$$\log p(\mathbf{Y}, \boldsymbol{\gamma} | \mathbf{D}, \boldsymbol{\mu}) = \sum_{v=1}^V \left[\log p(\boldsymbol{\gamma}_v | \boldsymbol{\mu}_v, \boldsymbol{\Sigma}) + \sum_{i=1}^N \log p(y_{vi} | d_{vi}, f(\gamma_{vi})) \right] \quad (6)$$

We can marginalize over $\boldsymbol{\gamma}_v$ to retrieve the **marginal likelihood**

$$\begin{aligned} p(\mathbf{Y} | \mathbf{D}, \boldsymbol{\mu}) &= \int p(\mathbf{Y}, \boldsymbol{\gamma} | \mathbf{D}, \boldsymbol{\mu}) d\boldsymbol{\gamma} \\ &= \prod_{v=1}^V \int p(\boldsymbol{\gamma}_v | \boldsymbol{\mu}_v, \boldsymbol{\Sigma}) \prod_{i=1}^N p(y_{vi} | d_{vi}, f(\gamma_{vi})) d\boldsymbol{\gamma}_v \end{aligned} \quad (7)$$

as well as the **marginal log-likelihood**

$$\begin{aligned} \log p(\mathbf{Y} | \mathbf{D}, \boldsymbol{\mu}) &= \log \int p(\mathbf{Y}, \boldsymbol{\gamma} | \mathbf{D}, \boldsymbol{\mu}) d\boldsymbol{\gamma} \\ &= \sum_{v=1}^V \log \int p(\boldsymbol{\gamma}_v | \boldsymbol{\mu}_v, \boldsymbol{\Sigma}) \prod_{i=1}^N p(y_{vi} | d_{vi}, f(\gamma_{vi})) d\boldsymbol{\gamma}_v \end{aligned} \quad (8)$$

2.3 Marginal log-likelihood estimation via Laplace approximation

The marginal log-likelihood reported above is not directly computable. We can use the Laplace approximation method to retrieve an approximation.

We first need to find a (local) maximum of the function to approximate through iterative numerical optimization, and using this, build a local Gaussian approximation by evaluating the matrix of second derivatives of the negative marginal log-likelihood [Bishop, 2007].

2.3.1 Laplace approximation

Assume we want to approximate a probability density $p(\mathbf{z}) = \frac{1}{Z} f(\mathbf{z})$, with $Z = \int f(\mathbf{z}) d\mathbf{z}$ being the normalizing coefficient. The Laplace method finds a Gaussian approximation $q(\mathbf{z})$ of $p(\mathbf{z})$ centered on a mode $\mathbf{z}_0$ of $p(\mathbf{z})$, such that $q(\mathbf{z}) = \mathcal{N}(\mathbf{z} | \boldsymbol{\mu} = \mathbf{z}_0, \boldsymbol{\Sigma} = \mathbf{H}^{-1})$, with $\mathbf{H}$ being the $N \times N$ Hessian matrix defined as

$$\mathbf{H} = -\nabla \nabla \log f(\mathbf{z})|_{\mathbf{z}=\mathbf{z}_0} \quad (9)$$

and $\mathbf{z}_0$ being a mode of $p(\mathbf{z})$

$$\left. \frac{df(\mathbf{z})}{d\mathbf{z}} \right|_{\mathbf{z}=\mathbf{z}_0} = 0 \quad (10)$$

The Laplace's approximation of the un-normalized log probability density is

$$\log f(\mathbf{z}) \simeq \log f(\mathbf{z}_0) - \frac{1}{2}(\mathbf{z} - \mathbf{z}_0)^T \mathbf{H}(\mathbf{z} - \mathbf{z}_0) \quad (11)$$

We take the exponential to obtain the estimate of the un-normalized distribution

$$f(\mathbf{z}) \simeq f(\mathbf{z}_0) \exp \left\{ -\frac{1}{2}(\mathbf{z} - \mathbf{z}_0)^T \mathbf{H}(\mathbf{z} - \mathbf{z}_0) \right\} \quad (12)$$

Using 12, we can approximate the normalizing constant Z

$$\begin{aligned}
Z &= \int f(\mathbf{z}) d\mathbf{z} \\
&\simeq f(\mathbf{z}_0) \int \exp\left\{-\frac{1}{2}(\mathbf{z} - \mathbf{z}_0)^T \mathbf{H}(\mathbf{z} - \mathbf{z}_0)\right\} \\
&= f(\mathbf{z}_0) \frac{(2\pi)^{N/2}}{|\mathbf{H}|^{1/2}}
\end{aligned} \tag{13}$$

Following previous equations, we can approximate the marginal log-likelihood $p(\mathcal{D}) = \int p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta})d\boldsymbol{\theta}$. Consider $\boldsymbol{\theta} = \mathbf{z}$, $f(\mathbf{z}) = p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta})$ and $Z = p(\mathcal{D})$, apply equation 13 and log-transform to obtain the following

$$\log p(\mathcal{D}) \simeq \log p(\mathcal{D}|\hat{\boldsymbol{\theta}}) + \log p(\hat{\boldsymbol{\theta}}) + \frac{N}{2}\log(2\pi) - \frac{1}{2}\log|\mathbf{H}| \tag{14}$$

In this case we can define the mode of $p(\boldsymbol{\theta}|\mathcal{D})$ as

$$\left. \frac{dp(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{d\boldsymbol{\theta}} \right|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}} = 0 \tag{15}$$

and the Hessian matrix as

$$\mathbf{H} = -\nabla\nabla\log p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta})|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}} \tag{16}$$

2.3.2 Application of the Laplace's approximation to our model

Let's apply equation 14 to our model

$$\log p(\mathbf{y}_v|\mathbf{d}_v, \boldsymbol{\mu}_v, \boldsymbol{\Sigma}) \simeq \log p(\mathbf{y}_v|\hat{\boldsymbol{\gamma}}_v; \mathbf{d}_v) + \log p(\hat{\boldsymbol{\gamma}}_v|\boldsymbol{\mu}_v, \boldsymbol{\Sigma}) + \frac{N}{2}\log(2\pi) - \frac{1}{2}\log|\mathbf{H}| \tag{17}$$

with

$$\begin{aligned}
\log p(\mathbf{y}_v|\hat{\boldsymbol{\gamma}}_v; \mathbf{d}_v) &= \sum_{i=1}^N \log \text{Binomial}(y_{vi}|\mathbf{d}_{vi}, g(\hat{\gamma}_{vi})) \\
&= \sum_{i=1}^N \log \binom{d_{vi}}{y_{vi}} + y_{vi}\log g(\hat{\gamma}_{vi}) + (d_{vi} - y_{vi})\log(1 - g(\hat{\gamma}_{vi}))
\end{aligned} \tag{18}$$

and

$$\begin{aligned}
\log p(\hat{\boldsymbol{\gamma}}_v|\boldsymbol{\mu}_v, \boldsymbol{\Sigma}) &= \log \text{Normal}(\hat{\boldsymbol{\gamma}}_v|\boldsymbol{\mu}_v, \boldsymbol{\Sigma}) \\
&= -\frac{1}{2}\log|\boldsymbol{\Sigma}| + \frac{N}{2}\log(2\pi) + (\hat{\boldsymbol{\gamma}}_v - \boldsymbol{\mu}_v)^T \boldsymbol{\Sigma}^{-1} (\hat{\boldsymbol{\gamma}}_v - \boldsymbol{\mu}_v)
\end{aligned} \tag{19}$$

In order to find the mode $\hat{\boldsymbol{\gamma}}$

$$\left. \frac{d \log p(\mathbf{y}_v|\boldsymbol{\gamma}_v; \mathbf{d}_v) + \log p(\boldsymbol{\gamma}_v|\boldsymbol{\mu}_v, \boldsymbol{\Sigma})}{d\boldsymbol{\gamma}_v} \right|_{\boldsymbol{\gamma}_v=\hat{\boldsymbol{\gamma}}_v} = 0 \tag{20}$$

and the Hessian matrix

$$\mathbf{H} = -\nabla\nabla\log p(\mathbf{y}_v|\boldsymbol{\gamma}_v; \mathbf{d}_v) + \log p(\boldsymbol{\gamma}_v|\boldsymbol{\mu}_v, \boldsymbol{\Sigma})|_{\boldsymbol{\gamma}_v=\hat{\boldsymbol{\gamma}}_v} \tag{21}$$

$\hat{\boldsymbol{\gamma}}$ is found with the limited-memory Broyden–Fletcher–Goldfarb–Shanno algorithm (L-BFGS).

2.4 Inference steps

The general steps to train the model and optimize the unknown parameters are the following

1. Compute the $N \times N$ kernel Σ , its inverse and log-determinant
2. Initialize the parameters (μ) and the optimizer (e.g. Adam optimizer)
3. For each iteration of the inference:

For each mutation v

Compute the Laplace approximation of the marginal likelihood $\log p(\mathbf{y}_v | \mathbf{d}_v, \mu_v, \Sigma)$

The steps to compute the Laplace approximation of the marginal likelihood (equation 17) are the following

1. Compute $\hat{\gamma}_v$ (equation 20) – Adam optimizer
2. Compute the Hessian matrix (equation 21) and its determinant
3. Compute the log-likelihood (equation 18) and log-prior (equation 19)
4. Compute the Laplace approximation (equation 17)

3 Latent Gaussian process with Gaussian approximation to Binomial likelihood

When the number of cells N is high, we can approximate the Binomial likelihood in the model explained above to a Gaussian distribution. The model would become as follows

$$y_{vi} \sim \text{Normal}(d_{vi}\theta_{vi}, d_{vi}\theta_{vi}(1 - \theta_{vi})), \quad \text{with } \theta_{vi} = f(\gamma_{vi}) = \frac{1}{1 + e^{-\theta_{vi}}} \quad (22)$$

$$\gamma_v \sim \text{Normal}(\mu_v, \Sigma) \quad (23)$$

Since there is the logit link for γ , this is not a conjugate Gaussian setting and the posterior is not Gaussian - more similar to a logistic regression framework.

Examples and explanations can be found here

- [Bishop, 2007]: section 4.5 about Bayesian logistic regression
- [Gelman et al., 2013]: section 13.5 about marginal posterior approximations and 21.3 about latent Gaussian process models
- [Rasmussen and Williams, 2006]: section 3.3 on Gaussian process classification

References

- [Bishop, 2007] Bishop, C. M. (2007). *Pattern Recognition and Machine Learning (Information Science and Statistics)*. Springer, 1 edition.
- [Gelman et al., 2013] Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. (2013). *Bayesian Data Analysis*. Chapman Hall/CRC Texts in Statistical Science Series. CRC, Boca Raton, Florida, third edition.
- [Rasmussen and Williams, 2006] Rasmussen, C. E. and Williams, C. K. I. (2006). *Gaussian Processes for Machine Learning*. The MIT Press.
- [Revell, 2012] Revell, L. J. (2012). phytools: An R package for phylogenetic comparative biology (and other things). *Methods in Ecology and Evolution*, 3:217–223.